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Potential risks of generative artificial intelligence integration into K-12 education: A scoping review

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ABSTRACT

The rapid integration of Generative AI (GenAI) into K-12 education has outpaced our understanding of its potential risks and unintended consequences. Existing reviews have prioritized higher education and technical promise, overlooking the potential developmental risks it poses to children and adolescents. This scoping review synthesizes 22 empirical studies from K-12 contexts to map the types of risks and concerns reported in research, compile mitigation strategies, and identify priority gaps that could guide more developmentally informed future studies. Across the included studies, reported risks clustered into three domains: (a) risks to psychological wellbeing, with evidence of emotional disconnection and social isolation; (b) risks to intellectual agency, comprising cognitive dependency, distorted self-assessment, and the erosion of creative authorship; and (c) risks to ecological environments, including limited institutional readiness, unclear governance, equity gaps, and privacy concerns that complicate safe and consistent student engagement with GenAI. Promising mitigation strategies identified include designing tasks that value process over product, using GenAI to generate scaffolding (hints) rather than direct solutions, and embedding tools within critical AI literacy curricula. Ultimately, this review suggests that without intentional, developmentally responsive governance, GenAI risks displacing the productive struggle and authentic expression necessary for learning and identity formation. We conclude that safe integration requires shifting the focus from technical adoption to ecological protection, ensuring that tools function as transparent scaffolds for human cognition rather than opaque substitutes for student agency in the K-12 context.

1. Introduction

The rapid evolution of artificial intelligence (AI), particularly generative AI (GenAI) powered by large language models (LLMs) such as ChatGPT, has transformed the educational landscape since its public debut in November 2022 (Allison et al., 2025; Chiu, 2024; Tu & Lu, 2025). GenAI systems can generate human-like text, sustain multi-turn dialogues, translate languages, and produce multimodal creative content on demand, enabling personalized instruction, enhanced engagement, and streamlined preparation for teaching and learning in education settings (Alfarwan, 2025; Labadze et al., 2023; Yusuf et al., 2024). Unlike earlier machine-learning tools that primarily delivered static content or operated as background analytics, contemporary GenAI applications intervene directly in students' meaning-making: they answer questions, model solutions, and co-produce artefacts within the flow of classroom and homework tasks (Chiu, 2024; Lai & Tu, 2024; S. Wang, Wang, Zhu, et al., 2024). This shift has made GenAI one of the most intensively discussed educational technologies in recent years, but it also raises a deeper, still insufficiently addressed question: *What happens to students' learning, self-expression, and help seeking if they rely on GenAI during the school years, while their thinking and social emotional skills are still developing*?

This question is especially critical because childhood and adolescence constitute sensitive periods of neurocognitive and psychosocial maturation (Best & Miller, 2010; Blakemore & Mills, 2014). Executive functions such as working memory, inhibitory control, and cognitive

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flexibility undergo substantial refinement across the school years and provide the foundation for self-regulated learning and higher-order reasoning (Best & Miller, 2010; Diamond, 2013). At the same time, students are forming beliefs about effort and ability, negotiating academic and creative identities, and learning to manage stress, relationships, and emotions in school contexts (Durlak et al., 2011; Macnamara & Burgoyne, 2023; Meeus et al., 1999; van der Zanden et al., 2020). Within this developmental backdrop, the continual availability of GenAI as an "always-on" problem-solver and co-author may encourage cognitive offloading, reduce opportunities for productive struggle and creative exploration, and blur boundaries between students' own contributions and AI-generated output (Dergaa et al., 2024; Zhai et al., 2024). Such patterns may, in turn, shape intrinsic motivation, self-efficacy, and perceived authenticity in ways that have implications for both learning and mental well-being, as students renegotiate what it means to "do the work themselves" in an era of ubiquitous AI support (Higgs & Stornaiuolo, 2024; Urmeneta & Romero, 2025).

Concurrently, GenAI has been rapidly embedded across multiple layers of the K-12 educational ecosystem, with clear benefits reported. Teachers report using GenAI to draft lesson materials, examples, quizzes, rubrics, and multimodal resources; to provide dialogic explanations, hints, and iterative feedback that support revision; and to assist with assessment design, item generation, and process-oriented review under explicit human oversight (Chiu et al., 2023; S. Wang, Wang, Zhu, et al., 2024). Beyond classroom instruction, affective computing applications analyze learner emotions to tailor support and inform special education, while institutions deploy GenAI in admissions, scheduling, analytics, and outcome prediction to increase responsiveness and efficiency at scale (Baidoo-Anu & Owusu Ansah, 2023; Mallik & Gangopadhyay, 2023; Vistorte et al., 2024). K–12 AI literacy initiatives further introduce students to AI concepts and ethics through project-based, game-based, and human–AI collaboration approaches using tools such as Teachable Machine, ML for Kids and Scratch, with promising outcomes for engagement, AI literacy, and problem-solving skills (Kong et al., 2024; Lee & Kwon, 2024; Yim & Su, 2025). Across this literature, benefits are consistently reported in terms of personalization, scalable formative feedback, and educator efficiency, with positive downstream effects on skills and achievement when GenAI is anchored in sound pedagogy and strong human oversight (Baidoo-Anu & Owusu Ansah, 2023; Chiu et al., 2023; Lee & Kwon, 2024).

However, a small but growing empirical base is beginning to show that GenAI may introduce distinctive risks for K–12 learners. In a large randomized trial in secondary mathematics, students using GPT-based tutors performed worse on follow-up exams without AI support, showed no long-term learning gains, and substantially overestimated their preparedness, consistent with a "learning penalty" driven by cognitive offloading and an inflated sense of mastery (Bastani et al., 2025). Qualitative work on AI chatbots used for emotional support among adolescents has documented perceptions of responses as generic or "cold," difficulties in handling complex emotions, privacy and manipulation concerns, and fears that over-reliance on chatbots could deter help-seeking from trusted adults (Chan, 2025). Studies with K–12 teachers highlight worries about reduced critical thinking, increased cheating and dependency, diminished human interaction and student–teacher bonding, educator stress under unclear policies, and equity concerns linked to differential access and AI bias (Cheah et al., 2025; Delello et al., 2025). Although existing studies mostly capture proximal outcomes, these may be precursors to long-term developmental consequences if they are not addressed.

Despite a surge in reviews on GenAI in education, current syntheses are only partially aligned with these developmental concerns. Existing reviews predominantly highlight GenAI's potential for personalized learning, scalable feedback, and teacher efficiency; Alternatively, they focus on technical flaws, infrastructural barriers and compliance issues, such as hallucinations, data bias, academic integrity and privacy (Alfarwan, 2025; Baidoo-Anu & Owusu Ansah, 2023; Mittal et al., 2024; S. Wang, Wang, Zhu, et al., 2024). When risks are mentioned, they are often framed in abstract terms such as "over-reliance" or "skill loss," without explicit linkage to cognition and socioemotional development, or creative identity in K-12 learners (Noroozi et al., 2024; Topali et al., 2025). Moreover, empirical work and debate on GenAI are heavily concentrated in higher education, with relatively less research and discussion focused on K-12 (Alfarwan, 2025; Mustafa et al., 2024; Topali et al., 2025). This gap is particularly consequential because early, unguided patterns of GenAI use during childhood and adolescence can plausibly crystallize into durable habits and dispositions toward effort, independence, and creative risk-taking (Bastani et al., 2025; Crone & Konijn, 2018; Klarin et al., 2024). In this context, there is a clear need to systematically map how GenAI-related risks for K–12 learners are currently conceptualized, operationalized, and studied.

The present scoping review responds to this need by systematically identifying and synthesizing empirical studies that report risks, challenges, and concerns of GenAI use among K–12 students, or among teachers whose practices shape students' learning experiences. It addresses three research questions:

1. How does the empirical literature on GenAI usage among K–12 students characterize the associated risks?
2. What pedagogical, policy, or design suggestions have been proposed to mitigate these risks?
3. What gaps and future directions emerge from the current evidence base that should guide the developmentally informed integration of GenAI into K–12 education?

2. Methods

2.1. Searching strategy

This scoping review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) guidelines (Tricco et al., 2018). A protocol was registered on the Open Science Framework (OSF; https://osf.io/cbamh) to ensure transparency and reproducibility. Consistent with scoping review methodology, no formal quality appraisal was conducted, as the aim was to map the breadth of evidence rather than assess study quality (Khalil et al., 2021).

A systematic search was conducted across four major electronic databases: Web of Science, Scopus, APA PsycInfo and ERIC (via EBSCOhost). An initial search was executed in July 2025, followed by an updated search in January 2026, to capture the most recent literature. The review timeframe was delimited to publications from January 1, 2022, through January 25, 2026, purposefully targeting empirical evidence emerging after the public release and widespread adoption of generative AI models (e.g., ChatGPT).

The search query combined three groups of keywords using the "AND" operator: AI-related terms ("generative AI," "large language models," "LLMs," "ChatGPT," "AI chatbots"); K-12 context terms ("K-12," "primary education," "secondary education," "high school," "classroom," "student," "adolescent"); Negative impact terms ("mental health," "anxiety," "dependence," "over-reliance," "critical thinking," "creativity," "cognitive development"). Full and detailed query strings used for each database are presented in Appendix I to ensure reproducibility. Additional records were identified through hand-searching reference lists of included studies and forward citation tracking via Google Scholar.

2.2. Eligibility criteria

Studies were selected based on a predefined set of eligibility criteria. The inclusion criteria are: (a) peer-reviewed empirical studies (quantitative, qualitative, or mixed-methods) published in English; (b) studies focusing on the use of Gen AI within a K-12 educational context; and (c) research that explicitly or implicitly reports the negative effects, challenges or risks of GenAI on students' outcomes, and which specifically addresses student-level issues rather than issues related to fine-tuned large models or teachers.

The exclusion criteria are: (a) studies published in languages other than English or non-research articles (e.g., news, commentaries, reviews); (b) studies exclusively emphasizing positive impacts of GenAI without discussion of negatives (c) research conducted outside K-12 settings (e.g., higher education, adult learning); or focusing on the non-direct use of GenAI technologies (e.g., traditional ML algorithms, robotics without generative components, or embed GenAI in online learning platforms) (d) studies limited to academic integrity issues (e.g., plagiarism detection); or technical/model performance limitations (e.g., AI accuracy, model hallucinations, or evaluation of fine-tuned large language models' performance metrics) without direct ties to student psychological or developmental outcomes (e) investigations centered on systemic factors (e.g., teacher training, policy implementation); lacking explicit links to student-level impacts or outcomes (e.g., student learning, behavior, or wellbeing); (f) studies with insufficient methodological rigor, including those that do not report participant information or involve teachers lacking documented teaching experience (e.g., pre-service teachers only).

2.3. Data extraction and synthesis

A structured data extraction form was developed a priori to ensure consistent charting of key study characteristics. For each included study, the following information was extracted: author(s) and year of publication, country or region, participant type and characteristics, sample size, study design, GenAI tools examined, research focus (i.e., subject area or educational context) and outcome measures (i.e., reported negative effects or risks). Where available, the form also captured any pedagogical, policy, or design recommendations proposed to mitigate the identified risks (i.e., practical recommendations).

Data extraction was conducted by one reviewer and independently checked for accuracy and completeness by a second reviewer; discrepancies were discussed until consensus was reached. Given the methodological heterogeneity of the included studies (spanning qualitative interviews, surveys, quasi-experiments, mixed-methods designs, and one randomized controlled trial), a narrative synthesis approach was adopted (Levac et al., 2010). Extracted data were first tabulated to provide an overview of study characteristics (e.g., geographical

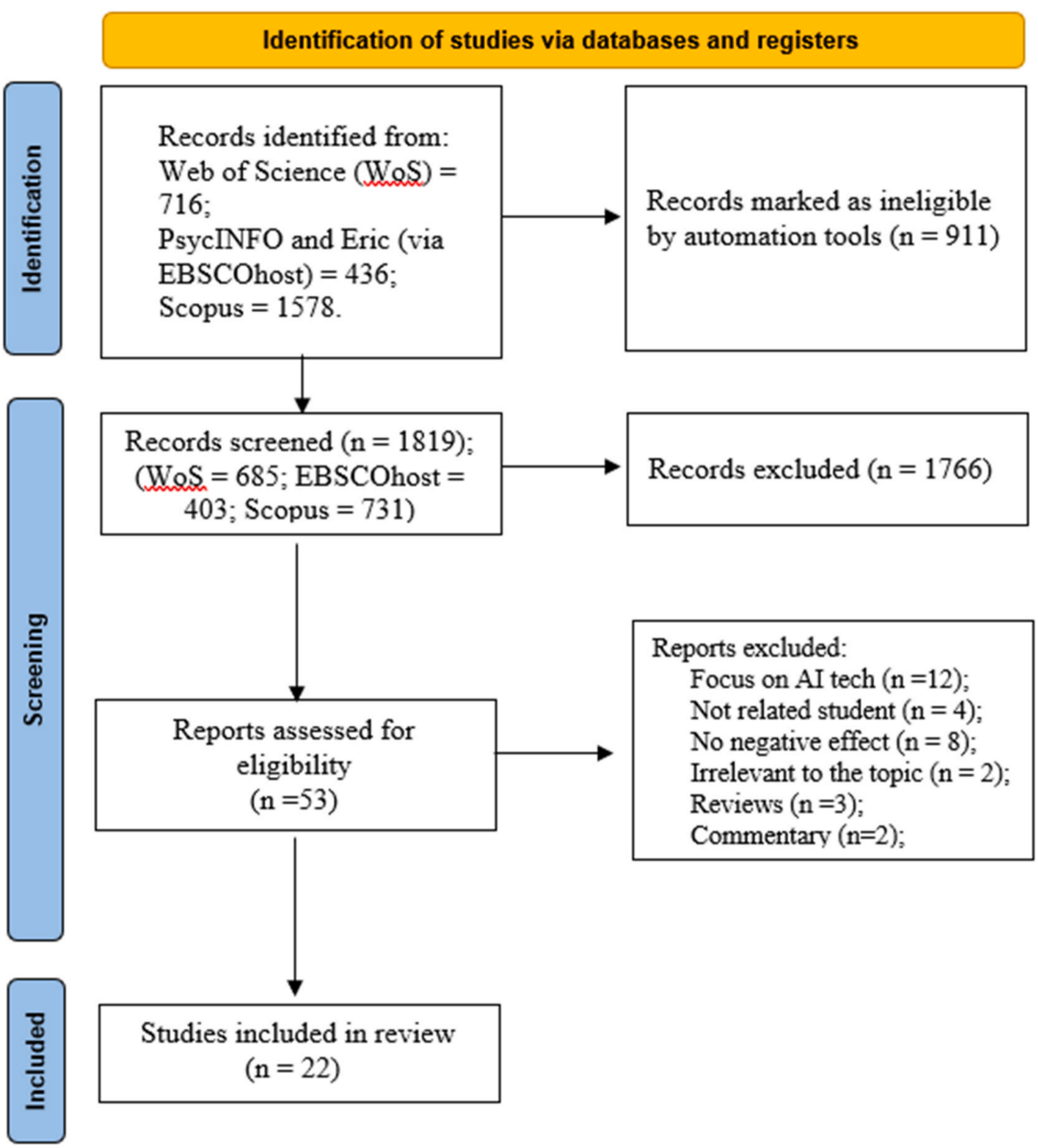

Fig. 1. PRISMA flow diagram.

distribution, participant types, GenAI tools, and outcome domains). Next, findings were inductively coded and grouped into thematic categories aligned with the review's primary and secondary research questions, focusing on (a) types and manifestations of risks, (b) proposed mitigation strategies in pedagogy, policy, and design, and (c) identified evidence gaps. Simple quantitative summaries (e.g., counts and proportions of studies reporting particular themes or outcome domains) were used to indicate the prevalence of concerns, while qualitative data were synthesized to illuminate contextual nuances and mechanisms.

3. Results

3.1. Study selection and overall characteristics

The database search and screening process identified 22 empirical studies that met all inclusion criteria, encompassing work published between 2024 and 2025. As summarized in Fig. 1 and Table 1, the studies covered a diverse set of educational systems, including the United States (4 studies), Greater China regions (Chinese mainland, Hong Kong, and Taiwan; 7 studies), Turkey (2 studies), Sweden (2 studies), and single studies from Iran, Spain, South Korea, Poland, Greece, and Kazakhstan, alongside one multi-country investigation. Research designs were heterogeneous, comprising qualitative studies ($n = 8$), quantitative survey or correlational studies ($n = 4$), mixed-methods designs ($n = 7$), quasi-experimental interventions ($n = 2$), and one randomized controlled trial (RCT).

Participant groups included K-12 students from kindergarten to upper secondary school (ages approximately 5-19 years) and teachers whose practices directly shape student experiences, including in-service K-12 teachers and mixed educator samples. Across studies, the primary GenAI tools examined were ChatGPT and related GPT-based models, with some work including other chatbots or multimodal systems (e.g., Gemini, MathGPT and Snapchat My-AI). The outcome domains are aligned with the focus of this review on psychological wellbeing (e.g., anxiety, emotional reliance and perceived isolation), intellectual agency (e.g., executive function, critical thinking, performance in exams, self-efficacy, originality and creative satisfaction) and concerns within educational ecosystems.

3.2. Risks of GenAI on K–12 students

3.2.1. Risks to psychosocial wellbeing

Evidence regarding the impact of GenAI on student emotional and social development reveals concerns about the quality of human-computer interaction and its potential to displace authentic social bonds. Findings in this domain cluster around two primary tensions: the emotional inadequacy of AI companions and the risk of social isolation.

First, student-level data highlight an emotional disconnect when using GenAI for support. In the study by Chan (2025), adolescents who used AI chatbots for mental health support explicitly described the interaction as “cold” and lacking the “human qualities” necessary for genuine empathy. These students reported trust issues, noting that the algorithmic responses felt generic and incapable of handling complex emotional situations. This frustration with the “robotic” or “weird” nature of AI outputs was also documented by LaMear and von Gillern (2025), where students struggled to craft prompts to communicate with GenAI, leading to frustration when GenAI misunderstood their vision. Furthermore, Higgs and Stornaiuolo (2024) found that this lack of authenticity contributed to a broader existential anxiety about the future value of human skills in an automated world.

Beyond emotional inadequacy, student data points to a displacement of social interaction. Yang et al. (2025) found that students utilizing ChatGPT experienced reduced interaction with teachers and peers, which directly contributed to lower engagement, fewer peer collaboration and a disrupted sense of “flow” in their learning. According to Rahimi and Teimouri (2025), using ChatGPT in a high school English class reduces genuine cultural exchange compared to a traditional class. Students in the study by Ergene and Ergene (2025) similarly expressed a preference for “human centered” learning, such as video or audio explanations from a teacher, noting that text only AI solutions failed to provide the necessary social scaffolding for their understanding.

Teachers corroborate these student reports, viewing social isolation as a critical structural risk. Delello et al. (2025) and Wang et al. (2025) indicated that a primary educator concern was that over-reliance on AI would reduce face to face interaction, potentially weakening the essential teacher-student relationship. This fear was echoed by Cheah et al. (2025), where teachers identified the “reduction of human interaction” as a significant pedagogical barrier to GenAI adoption. Furthermore, Jang and Choi (2025) noted that this shift in interaction patterns might diminish teacher-student relationship and teacher authority, as students could begin to view the AI as a more reliable or accessible source of support than their human instructors.

3.2.2. Risks to intellectual agency

Direct empirical evidence from student data indicates that unguided GenAI use is negatively related to cognitive function skills and creative ownership. In the cognitive domain, evidence suggests that reliance on GenAI can hinder independent performance and distort self-assessment. A randomized controlled trial by Bastani et al. (2025) demonstrated that students using a basic GPT tutor performed 17% worse on subsequent non-AI exams compared to their peers, indicating that the tool functioned as a ‘crutch’ rather than a scaffold. This dependency appears particularly pronounced among vulnerable groups: Klarin et al. (2024) observed that adolescents with executive function challenges perceived AI as significantly more useful and were more likely to use it for schoolwork. This suggests a tendency to over-rely on AI to compensate for cognitive deficits.

Beyond performance metrics, GenAI integration is also related to students’ metacognitive calibration and self-efficacy. Despite lower objective performance, students in the Bastani et al. (2025) study overestimated their abilities. This misalignment is reinforced by findings from Yang et al. (2025) and Chan and Lee (2025), who report that students often attribute their success to the AI rather than their own effort, a pattern indicative of reduced learning self-efficacy.

Furthermore, the increased availability of AI tools has been associated with reduced cognitive engagement and critical thinking. Students often “outsource” analytical tasks (Kilde-Westberg et al., 2025; Zhu et al., 2025) or become distracted when unsupervised (Li et al., 2025). Disturbingly, this may foster a “ready-made” mentality: Ergene and Ergene (2025) and Raimkulova et al. (2025) found that, even when encountering frequent inaccuracies in models such as Gemini and GPT-3.5, students rarely attempted to verify the information, instead opting for passive acceptance.

In the creative domain, the integration of GenAI may erose student voice and diminish authorship control. For primary school students, GenAI could act as an intrusive force rather than a supportive scaffold. LaMear and von Gillern (2025) observed that primary school students found AI-generated text ‘mechanistic’ or distorted, overriding their original creative vision rather than collaborating with it. This friction is exacerbated by the perceived perfection of algorithmic output. Song et al. (2025) observed that AI-generated stories were deemed “too perfect”, thereby limiting the co-construction process in primary writing classes and potentially displacing student contributions.

Among secondary students, this displacement manifests as a crisis of authenticity. Higgs and Stornaiuolo (2024) found that students struggled to distinguish their own contributions from algorithmic outputs, leading to a pervasive sense that the final product was ‘inauthentic’ or ‘not really mine.’ This alienation is particularly acute among students with high aesthetic interests, whom Skop and Frania (2024) found to perceive AI as a constraint on their ability to produce uniquely personal work. Ultimately, these limitations appear to hinder creative development and exacerbate the achievement gap. Chu et al. (2025) reported no

Table 1
Key characteristics of the reviewed studies.

First author (Year), Country/ Region	Participant Type	Participant Characteristics	Research Focus	AI Tool(s) and its roles	Research Design	Key Findings (Risks)	Practical Recommendations
Bastani et al. (2025), Turkey	Student	N ~ 1000 students across grades 9–11 in a high school	GPT-4 based tutors on student performance and skill learning in a high school math class	Tools: GPT-4 Roles: Math tutor; question responder; hint/ feedback provider.	Experimental (randomized controlled trial)	- Students using AI tutors performed 17% worse on follow-up exams. - No long-term learning gains. - Students using AI tutors overestimated their learning despite lower outcomes.	- Build safeguards against over-reliance. - Design AI to offer stepwise hints rather than direct answers. - Prioritize designs that support durable learning and retention. - Deploy cautiously.
Chan and Lee (2025), Hong Kong SAR	Student	N = 5 secondary school students (ages 14–17)	Use of GenAI tools in reflective writing	Tools: GenAI tools Roles: Idea generator, proofreader, reflection cue provider.	Qualitative (case study)	- Over-reliance led to "AI guilt" and reduced self-efficacy. - AI text was often shallow, weakening deep reflection. - Risk of losing personal voice and authenticity. - Fear that reflections would become homogenized.	- Provide explicit guidance on ethical GenAI use; emphasize personal reflection. - Design rubrics/ tasks to reward depth, originality, and insight rather than surface polish. - Teach AI literacy alongside "reflection literacy" (critically evaluate limits). - Position teachers as mentors balancing AI support with independent thinking.
Chan (2025), Hong Kong SAR	Student	N = 69 secondary school students (ages 13–17)	Use of GenAI chatbots as therapeutic tools in school-based mental health services	Tools: AI chatbots Roles: Mental health support; therapeutic conversation simulator.	Qualitative (thematic analysis of written reflections)	- Limited empathy and therapeutic depth; responses felt "cold/ generic." - Difficulty understanding complex emotions or situations. - Trust issues regarding privacy and manipulation. - Over-reliance may deter students from seeking human help.	- Use only as a supplementary tool with human oversight. - Improve AI's empathetic capabilities. - Establish clear institutional policies on privacy, ethics, and boundaries. - Provide AI literacy training for students and counselors.
Cheah et al. (2025), USA	Teacher	N = 89 K-12 teachers (age unknown)	Teachers' preparedness, usage, and barriers to GenAI integration	Tools: ChatGPT, etc. Roles: Lesson prep; assessment generator; admin support.	Mixed methods (cross-sectional survey with closed and open-ended questions)	– 61% teachers felt unprepared; 44% actively used GenAI. - A major obstacle was the lack of institutional guidance. - Other barriers included ethical concerns, fear of reduced human interaction, doubts about AI's capabilities, and privacy concerns.	- Develop flexible, credit-eligible training (especially for rural teachers). - Address teacher beliefs and attitudes alongside technical skills. - Promote AI literacy (including ethics) for educators and students. - Provide actionable school policies and strategic support. - Foster partnerships with tech providers to ensure equitable access and safe integration
Chu et al. (2025), Taiwan	Student	N = 60 high school students (mean age 16);	Gen AI on students' learning achievement and creativity in ancient poetry course	Tools: Copilot Roles: Image generator; visualizer for poetry contexts.	Experimental (quasi-experimental)	- The GenAI group showed better learning achievement. - Students with high creative tendency in	- Pair GenAI drawing with collaborative support so low-creativity learners benefit from peers.

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Table 1 (*continued*)

First author (Year), Country/ Region	Participant Type	Participant Characteristics	Research Focus	AI Tool(s) and its roles	Research Design	Key Findings (Risks)	Practical Recommendations
						the GenAI group showed significantly improved self-efficacy. - No significant differences between the groups in overall creative tendency or learning motivation scores. - Low creativity/tool-skill students felt frustrated or inadequate.	- Provide time and step-by-step training to reduce cognitive load. - Use GenAI to offload drawing but require deep text interpretation. - Extend duration and integrate online interaction to build team creativity.
Delello et al. (2025)*, USA, Italy, & other international locations	Teacher	N = 353 educators across K-12 and higher education	Impact of generative AI on teaching, student learning, mental health	Tools: ChatGPT, etc. Roles: Teaching assistant; admin automator; tutor.	Mixed methods (cross-sectional survey with closed and open-ended questions)	- Concerns regarding plagiarism and reduced critical thinking. - Educator stress due to unclear policies and misuse detection. - Risk of reduced human interaction/ bonding. - Fears of dependency and inflated self-perception. - Equity, bias, and data-security risks.	- Develop clear institutional policies for staff and students. - Frame AI as a complement, not a replacement, for human instruction. - Provide AI literacy and ethical training for educators. - Redesign assessments to focus on critical thinking and personal expression.
Ergene et al. (2025), Turkey	Student; Teacher	N = 160 high school students (ages 15-18); N = 80 mathematics teachers (age unknown)	Use of GenAI tools to solve math problems	Tools: ChatGPT (GPT-4o, GPT-4, GPT-3.5), MathGPT, Gemini Roles: Math problem solver; step-by-step explainer.	Mixed methods (descriptive + individual interview)	- Participants noted potential for incorrect solutions and the risk of reducing student effort (e.g., homework integrity). - Some found detailed solutions “overly detailed” or “boring” depending on the learner's level - Technical constraints included the need for internet access and prompt engineering requirements.	- Use stronger models (e.g., GPT-4) as supplements, not replacements. - Teach ethical use and discourage passive copying. - Teach prompting skills for clearer, targeted responses. - Maintain hybrid learning with human-led discussion/instruction.
Higgs et al. (2024), USA	Student	N = 131 high school students (grades 11–12, age unknown); 12 of them participated in focus groups	Uses of and ethical concerns about GenAI in writing and human identity	Tools: Various (e.g., ChatGPT) Roles: Writing support; idea catalyst; routine task handler.	Mixed methods (cross-sectional survey + focus group)	- Concerns about authenticity, creativity, and emotional depth in AI writing. - Ethical uncertainty regarding plagiarism and fairness. - Awareness of AI bias reinforcing social inequities. - Anxiety about future implications for labor and truth.	- Promote sociotechnical literacy and critical classroom discussions. - Use inquiry-based learning that addresses both benefits and risks. - Center student voices in research, policy, and curriculum. - Address access and equity across socioeconomic backgrounds.
Jang et al. (2025)*, Korea	Teacher	N = 10 experienced physics educators (high school & university) with prior AI exposure	Korean physics educators' views on ChatGPT in education and its potential future impacts	Tools: ChatGPT-4 Roles: Physics problem solver; inquiry-based learning aid.	Qualitative (individual interview)	- Reliability and accuracy were inconsistent. - Over-reliance may weaken problem-solving ability. - Prompt-skill disparities may widen educational inequality.	- Shift focus to curiosity and higher-order thinking rather than answer-getting. - Emphasize student-centered learning and metacognitive development.

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Table 1 (*continued*)

First author (Year), Country/ Region	Participant Type	Participant Characteristics	Research Focus	AI Tool(s) and its roles	Research Design	Key Findings (Risks)	Practical Recommendations
						- Risks of academic dishonesty and diminished teacher authority.	- Assess thinking processes, not just outputs. - Improve teacher AI literacy and scientific communication. - Establish ethical AI use guidelines and professional learning communities for support.
Kilde-Westberg et al. (2025), Sweden	Student	N = 19 high school students (age unknown)	Use of ChatGPT as the lab partner during physics laboratory investigating acoustic levitation and speed of sound	Tools: GPT-3.5 Roles: Lab partner; physics tutor; data analysis guide.	Qualitative (case study)	- Students primarily used ChatGPT for conceptual explanations and formula retrieval. - Students with more prior physics knowledge could evaluate ChatGPT's answers more critically. - Students with low knowledge/literacy were least able to detect errors. - Students with less prior knowledge struggled to verify incorrect or hallucinated information, leading to misconceptions.	- Teach how GenAI works (training data, hallucinations, limits) and evaluation skills. - Design labs where students must compare, check, and justify AI outputs. - Actively monitor/ intervene to correct AI-induced misconceptions. - Use GenAI as a supplemental hint partner; keep core reasoning with students.
Kladaki et al. (2025), Greece	Teacher	N = 67 primary (38.8%) and secondary (61.2%) school teachers (age unknown)	Teachers' beliefs about the pedagogical use of ChatGPT in arts education	Tools: ChatGPT Roles: Arts creativity aid; script/poem generator; brainstorming.	Qualitative (individual interview)	- Risk of copying/ plagiarism and reduced genuine engagement. - Anticipated drop in critical thinking/ creativity; passivity; less authentic work. - Misinformation risks in nuanced artistic/ cultural contexts. - Concerns: skill erosion, standardization, reduced human role, copyright/data ethics.	- Provide targeted professional development on capabilities/limits and pedagogy for arts. - Establish clear frameworks for ethics, copyright, and data protection. - Improve infrastructure/ access for equity. - Scaffold as a supplementary tool (ideas/prompts) while protecting student voice.
Klarin et al. (2024), Sweden	Student	N = 385 (middle school, mean age 14); N = 359 (high school, mean age 17)	Relationship between executive functioning, academic outcomes, and GenAI usage	Tools: ChatGPT with other tools (e.g., Snapchat MY-AI, Youchat, Bing-chat, Socratic, etc.) Roles: Homework assistant; summarizer; text improver.	Quantitative (cross-sectional survey)	- Adolescents with more executive function problems were more likely to use AI and found AI more useful for schoolwork. - No significant associations between AI usage/perceived usefulness and academic achievement. - Boys were more likely to use ChatGPT than girls, though girls rated its usefulness slightly higher (not statistically significant).	- Monitor use among students with executive function problems to prevent substituting cognition. - Implement thoughtfully to support, not replace, learning. - Conduct further research on adolescent usage patterns.

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Table 1 (*continued*)

First author (Year), Country/ Region	Participant Type	Participant Characteristics	Research Focus	AI Tool(s) and its roles	Research Design	Key Findings (Risks)	Practical Recommendations
						- Younger students showed low adoption rates.	
LaMear et al. (2025), USA	Student; Teacher	N = 28 elementary students (Kindergarten to 5th grade); N = 4 teachers (age unknown)	Strengths and challenges of AI in the elementary writing classroom	Tools: ChatGPT Roles: Story brainstorming; feedback provider; image generator.	Qualitative (case study)	- Students struggled to craft prompts that yielded their desired output, leading to frustration when GenAI misunderstood their vision. - Students identified racial and gender biases in GenAI outputs. - GenAI produced errors, illegible text in images, or hallucinated information, prompting students to question its trustworthiness.	- Teach prompting strategies and ethical guidelines. - Use AI bias as a learning tool. - Balance technology with traditional creation methods.
Li et al. (2025), Taiwan	Student	N = 93 (11th grade students, mean age 16)	Effectiveness of a ChatGPT-AIPSS (Activate, Inquire & Participate, Share, and Summarize) learning approach in a Sustainable Development Goals course.	Tools: ChatGPT Roles: Learning support; critical thinking partner.	Experimental (quasi-experimental)	- Prone to distraction without supervision. - Risk of plagiarism. - Limited impact for students with low critical-thinking skills.	- Design differentiated support based on critical-thinking levels. - Provide closer guidance for struggling students. - Implement supervision mechanisms.
López Santos et al. (2025), Spain	Teacher	N = 379 secondary school teachers (age unknown)	Teachers' perceptions of ChatGPT's influence and integration in education	Tools: ChatGPT Roles: Content generator; planning aid; self-learning tool.	Quantitative (cross-sectional survey)	- Low adoption: 37% used it for teaching tasks, 15% used in class activities. - High concerns: loss of student creativity (80%), reduced information search/ analysis skills (65%), and plagiarism (approx. 40%). - A majority (68.1%) agreed that tasks need redesigning to prevent plagiarism. - Perceptions varied by specialty, age, and gender.	- Provide targeted training based on demographics and subjects. - Establish institutional policies and ethical standards. - Reform legislation to update academic integrity definitions.
Rahimi et al. (2025), Iran	Student	N = 236 high school students (221 male, 15 female; 97% aged 17–18)	How AI-assisted language learning help develop students' 21st-century digital skills	Tools: ChatGPT Roles: Language tutor; text generator; collaboration facilitator.	Quantitative (cross-sectional survey)	- AI's human characteristics was negatively linked to collaborative skills (encouraged individual use). - Digital self-authenticity negatively correlated with collaboration. - Inaccurate outputs caused confusion until teacher correction. - Reduced genuine cultural exchange compared to traditional classes.	- Design tasks balancing personalization with structured peer interaction (debates, projects). - Train students to verify outputs via peers/external sources. - Focus PD on using AI features to build 21st-century skills (not just language). - Update curricula to systematically include AI and digital skills.
Raimkulova et al. (2025), Kazakhstan	Student; Teacher	N = 119 third grade students (aged 8–9); N = 4 novice teachers	Effectiveness of GenAI- powered game-based instruction in	Tools: ChatGPT 4o mini Roles: Voice-game mediator; pronunciation	Mixed method (quasi-experimental and reflective journals)	- Grammar gains were statistically equivalent between the two groups.	- Train pre-service teachers on facilitation, prompting, and troubleshooting.

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Table 1 (*continued*)

First author (Year), Country/ Region	Participant Type	Participant Characteristics	Research Focus	AI Tool(s) and its roles	Research Design	Key Findings (Risks)	Practical Recommendations
			primary English class	feedback; anxiety reducer.		- Monitoring was hard due to private headphone interactions. - Students often requested translations rather than facing the challenge. - Vulnerable to connectivity issues and hidden prep workload.	- Use pre-piloted prompts and "tech-check" routines for stability. - Add complementary assessments (sharing/follow-ups) to evidence private learning. - Use selectively to manage workload and preserve novelty.
Skop et al. (2024), Poland	Student	N = 75 high school students (ages 15–19)	Disposition and knowledge towards the use of Gen AI	Tools: ChatGPT Roles: Writing tool; translator; problem solver.	Quantitative (diagnostic survey and psychological tests)	- Students with practical/aesthetic interests felt AI limited creativity. - Boys reported higher AI knowledge/ interest than girls. - Students acknowledged the potential for errors and cheating. - No strong link between personality traits and interest.	- Integrate where students see value (e.g., writing, problem solving). - Teach prompt literacy and responsible media education. - Address gender gaps with inclusive AI education. - Normalize guided classroom use to reduce informal misuse.
Song et al. (2025), USA	Student; Teacher	N = 79 5th-grade elementary students; N = 6 elementary teachers (age unknown)	Perceived educational benefits and challenges of GenAI-supported creative mathematical writing	Tools: GPT-4 and DALL·E 3 Roles: Math story generator; creative writing facilitator.	Qualitative (individual interview, open-ended survey, and classroom observation)	- Students often failed to critically evaluate/ revise AI stories. - Weak problem-generation competence; struggled to create story seeds. - AI stories were "too perfect," limiting co-construction. - Lack of pedagogical features (feedback) and social features (sharing). - Limited input modes (no speech-to-text) barred accessibility.	- Build in evaluation/ revision supports (checklists, editing prompts). - Calibrate AI to leave "deliberate gaps" for students to refine. - Add pedagogical functions (hints) and social features (sharing/ comments). - Scaffold autonomy and improve accessibility (multimodal input). - Provide subject-specific teacher professional development.
Wang et al. (2025), China	Student; Teacher	N = 132 K–6 students; N = 2 experienced K–6 art teachers (age unknown)	Use of GenAI for group project-based offline art courses in elementary schools	Tools: GPT-4 and DALL-E 3 Roles: Query assistant; project collaborator; image generator.	Qualitative (four-phase filed study)	- Students struggled to formulate queries, leading to frustration. - Misunderstood capabilities (e.g., asking DALL-E for text instructions). - Group work often excluded quieter students from GenAI interaction. - Teacher concern: copying short-circuits learning/original thinking. - High effort needed for productive use.	- Clarify roles via metaphors (e.g., GPT as "think tank," DALL-E as "artist"). - Scaffold query articulation (templates, examples). - Support tool chaining (text → image prompts). - Align GenAI use with task complexity/skill level. - Require documentation of prompts/outputs for assessment.
Yang et al. (2025), Taiwan	Student	N = 153 s-grade girls from an all-girls high school (age unknown)	Effects of ChatGPT in assisting high school programming learning	Tools: ChatGPT Role(s): a virtual teaching assistant in programming learning;	Mixed method (quasi-experimental and post-hoc interview)	- Students using ChatGPT reported lower flow experience, self-efficacy, and learning achievement.	- Use guidance practice transformation models to support reflective use.

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Table 1 (*continued*)

First author (Year), Country/Region	Participant Type	Participant Characteristics	Research Focus	AI Tool(s) and its roles	Research Design	Key Findings (Risks)	Practical Recommendations
				debugger; code verifier.		- Limited adaptability led to disengagement. - Over-attributing success to AI weakened student confidence. - Errors and unclear responses reduced learning clarity.	- Match AI integration to learners' cognitive/emotional readiness. - Use evaluative tasks to force critical critique of AI code. - Develop adaptive AI that meets diverse student needs.
Zhu et al. (2025), China	Student; Teacher	N = 127 middle school students (Grade 7; mean age ≈12.4); N = 1 teacher	Impact of assignment completion assisted by LLM chatbot on middle school student learning	Tools: LLM-based educational chatbot built on AliCloud Tongyi Qianwen Roles: Assignment assistant; emotional support; autonomy facilitator.	Mixed method (quasi-experimental and teacher reflection)	- Students in the LLM group gained significantly lower scores on materials refinement and knowledge tests. - student copy without evaluation led to superficial engagement. - High trust fostered passivity; skilled students dominated group work. - Typing/time constraints encouraged short prompts and shallow use.	- Redesign assessments for higher-order objectives. - Explicitly assess student evaluation and application of AI content. - Scaffold collaboration with structured group roles (e.g., checker). - Teach use as a “thinking assistant” alongside emotional support. - Develop policies and child-centered design (e.g., hallucination reduction).

Note. *Studies with mixed samples (K-12 and university).

overall increase in creative tendencies among students using GenAI in poetry courses; Low-creativity students found it difficult to use AI tools to effectively scaffold their learning, which led to feelings of frustration.

Teachers also identified “over-reliance” as the primary threat to student agency, though these findings represent anticipated concerns rather than measured outcomes. In the study by Delello et al. (2025), educators expressed fear that students would develop a “false sense of personal abilities” by bypassing the struggle of learning. This aligns with findings from Ergene and Ergene (2025) and Jang and Choi (2025), where teachers warned that students were losing independent problem-solving skills and becoming dependent on readymade answers. Teachers in Jang and Choi (2025) also noted that the hallucination problem placed a new, paradoxically high cognitive burden on students to verify information, which is a skill that educators felt most students had not yet developed.

3.2.3. *Risks to the ecological learning environment*

A consistent finding across studies focusing on teachers is the profound lack of systematic preparation. Cheah et al. (2025) reported that the majority of teachers (61%) felt “not at all” or only “slightly” prepared to use GenAI, identifying the absence of district level policies and goals as a primary barrier. This finding is reinforced by Delello et al. (2025), who noted that the lack of clear institutional governance was a major source of “technostress” for educators, who struggled to navigate the technology without strategic guidance. Similarly, López Santos et al. (2025) found that most surveyed teachers had not received specific training on GenAI tools, resulting in minimal practical integration in the classroom (15%). This widespread lack of pedagogical competence suggests that most students are currently navigating powerful AI tools without expert scaffolding, increasing the likelihood of the misuse patterns described in previous sections.

Teachers also voiced significant concerns regarding the ethical implications of GenAI deployment. Academic integrity was a dominant theme, with around 40% of teachers in the study by López Santos et al. (2025) explicitly viewing GenAI use as a form of plagiarism. Delello et al. (2025) further highlighted that teachers are increasingly forced to redesign assignments solely to detect or prevent AI-driven cheating. This reactive approach adds to their professional burden.

Beyond integrity, equity emerged as a critical implementation challenge. Jang and Choi (2025) warned that GenAI could exacerbate educational inequality, as the quality of AI output depends on prompting skills that may vary by student background, potentially widening the achievement gap between high and low performing groups. Furthermore, Kladaki et al. (2025) raised concerns that AI algorithms might reproduce racial disparities or lead to unjust accusations of dishonesty against specific student groups. Finally, Cheah et al. (2025) cited unresolved data privacy and security risks as significant barriers preventing widespread, safe adoption in schools.

3.3. *Suggested strategies to mitigate potential risks*

Across studies, proposed mitigation strategies focused on constraining how GenAI is positioned in learning, strengthening human oversight, and redesigning tools and assessments to promote active engagement.

Studies have recommended the use of chatbots as supplementary support for psychological wellbeing issues, within clearly defined boundaries and with adult supervision for emotionally charged topics; AI outputs should be used as prompts for classroom discussions about empathy, fairness and representation, rather than as standalone counsellors (Chan, 2025; Delello et al., 2025; LaMear & von Gillern, 2025).

To address cognitive risks, several studies stressed that GenAI should support, not replace, core cognitive processes. Suggested approaches included designing AI tutors that provide graded hints and metacognitive prompts rather than complete solutions, emphasizing process-oriented and performance-based assessments that require explanation and reflection, and implementing supervision and guidance for students with lower executive function skills who may be especially vulnerable to plagiarism and over-reliance (Bastani et al., 2025; Ergene & Ergene, 2025; Jang & Choi, 2025; Klarin et al., 2024; Li et al., 2025; Yang et al., 2025).

For creativity, authenticity, and identity, recommendations centered on balancing AI with traditional creation methods, explicitly teaching AI as a tool for refining rather than replacing students' ideas, and framing GenAI as a collaborator whose outputs are always subject to critical editing and personalization (LaMear & von Gillern, 2025; Skop & Frania, 2024).

At a broader systems level, multiple studies highlighted the need for institutional policies that clarify acceptable AI use, update definitions of academic integrity, and address data privacy and equity concerns; Studies also called for comprehensive, ongoing professional development to build teachers' AI literacy, with specific attention to the needs of rural educators, subject-specific applications, and demographic differences among students (Cheah et al., 2025; López Santos et al., 2025; Skop & Frania, 2024).

3.4. Research gaps and underexplored areas

Our review revealed some literature gaps. First, most projects were short-term classroom implementations, with only one randomized trial (Bastani et al., 2025) and a small number of quasi-experiments (Chu et al., 2025; Li et al., 2025; Raimkulova et al., 2025; Yang et al., 2025; Zhu et al., 2025), which limits inferences about how GenAI use shapes longer-term trajectories of psychological wellbeing, cognitive development, and creativity across schooling. Intervention durations were typically brief, spanning from three weeks (Raimkulova et al., 2025) to eight weeks (Li et al., 2025), thereby restricting our understanding of sustained developmental impacts. Second, the evidence is concentrated in a small set of educational systems (e.g., United States, Greater China, and a few European contexts), so cross-cultural, linguistic, and policy variation in GenAI's developmental impact remains largely unknown.

Third, while several studies raise concerns about biased outputs (LaMear & von Gillern, 2025; Wang et al., 2025) or differential access (Cheah et al., 2025), few empirically investigate how risks such as cognitive offloading, creative suppression, and emotional dependency are disproportionately borne by students from minoritized racial and linguistic backgrounds, low-income communities, or those with disabilities. The current literature alludes to the danger of widening educational gaps, but the specific mechanisms through which GenAI might deepen, rather than narrow, existing educational and wellbeing disparities remain a critical and underexplored area. Future work could move beyond documenting differential use and instead model how these tools interact with systemic inequities to shape developmental pathways.

Finally, the current literature focuses heavily on text-based tools in a narrow set of subjects (e.g., mathematics, programming, and writing) and only two studies directly compared alternative GenAI designs (e.g., direct-answer versus scaffolded-hint tutors or different chatbot models), indicating that many subject areas and design configurations relevant to K-12 development remain empirically untested (Bastani et al., 2025; Ergene & Ergene, 2025; Higgs & Stornaiuolo, 2024; LaMear & von Gillern, 2025; Li et al., 2025; Yang et al., 2025).

4. Discussion

This scoping review synthesized 22 empirical studies to map potential risks of GenAI in K-12 education, corresponding mitigation strategies and current literature gaps. The findings reveal three primary risk domains: (1) psychological wellbeing, characterized by emotional disconnection and social isolation; (2) intellectual agency, manifesting as reduced independent learning, inflated self-assessments, and weakened creative ownership; and (3) ecological environments, involving systemic issues such as limited institutional readiness, unclear governance, and equity concerns. These results align with the OECD (2026) report, confirming that while GenAI may enhance immediate task performance, it does not guarantee learning gains. Instead, offloading cognitive tasks to chatbots risks fostering metacognitive passivity and disengagement, potentially impeding long-term skill acquisition. However, given the small and heterogeneous evidence base and the absence of formal quality appraisal, our findings should be interpreted as an early map of reported concerns rather than estimates of prevalence, magnitude, or long-term developmental impact.

Across studies, the literature also described mitigation approaches and future directions, including using GenAI to generate hints, examples, and counterarguments while requiring students to show their reasoning, shifting assessment toward process focused tasks, such as in class explanations, and embedding GenAI within AI literacy and critical digital citizenship curricula that foreground bias, fairness, privacy, and authenticity.

4.1. Psychological wellbeing and relational dynamics

Findings related to the psychological wellbeing can be interpreted through several psychological frameworks. Students' descriptions of GenAI as a "cold," untrustworthy confidant and their reluctance to disclose sensitive information to chatbots highlight the enduring importance of secure, human relationships for emotional regulation and help-seeking, aligning with attachment and social support theories that emphasize the role of trusted others as a buffer against distress (Ainsworth & Bowlby, 1991; Chan, 2025). At the same time, evidence that academic stress and performance expectations mediate the relationship between self-efficacy and AI dependency in older students suggests that problematic GenAI use may function as a coping strategy within high-pressure environments rather than as a purely technological effect, echoing transactional stress–coping and compensatory technology use models (Lazarus, 1993; Zhang et al., 2024).

From a motivational perspective, concerns that GenAI can undermine students' meaningful contact with teachers and peers, and lead some adolescents to attribute success to AI rather than to their own efforts can be read through the lens of self-determination theory: over-reliance on GenAI risks eroding autonomy, competence, and relatedness, three basic psychological needs that underpin wellbeing and intrinsic motivation (Deci & Ryan, 2012). When GenAI is allowed to replace face-to-face interaction and authentic self-expression, it may also contribute to social displacement and feelings of inauthenticity, which are themselves linked to anxiety, isolation, and identity confusion during adolescence (see Kolding et al., 2024 for a review).

Human–computer interaction studies of mental-health chatbots show a similar pattern: people often turn to chatbots for quick reassurance or anonymous advice, but heavy reliance is linked to greater anxiety, frustration when the bot misses nuance, and delays in seeking help from trusted adults or professionals (Kolding et al., 2024). Taken together, this suggests that GenAI may intensify a broader shift toward technology-mediated support that is convenient yet emotionally thin, especially when it is framed as a replacement rather than a complement to human care (Chan, 2025; Cheah et al., 2025; Delello et al., 2025; Higgs & Stornaiuolo, 2024). To safeguard students' socioemotional development, schools therefore need clear guardrails on chatbot use and explicit boundaries between informational uses (such as explaining coping strategies) and any quasi-therapeutic roles. Structural changes are equally important, such as more manageable workloads, assessment and grading practices that do not reward the last-minute use of AI, and classroom norms that normalize asking humans for help. This will make GenAI less attractive as a way of coping or completing work.

4.2. Intellectual agency: cognitive development and learning

Concerns have been raised that GenAI may hinder cognitive development due to the risk of over-dependence and cognitive offloading (Ergene & Ergene, 2025; Klarin et al., 2024; N. Wang, Wang, Zhu, et al., 2024). Our review further identifies a developmental nuance: for primary school learners, reliance on GenAI may interfere with the "automation" of foundational skills such as syntax or calculation (LaMear & von Gillern, 2025); whereas for secondary adolescents, the risk shifts toward the erosion of self-efficacy and academic identity (e.g., Klarin et al., 2024).

From a cognitive perspective, evidence from experimental studies indicates that GenAI-supported practice can, under certain conditions, weaken subsequent unaided performance and foster overconfidence, a pattern that aligns more with maladaptive cognitive offloading than with effective scaffolding (Bastani et al., 2025; Yang et al., 2025). When GenAI supplies complete solutions on demand, it may reduce cognitive load but simultaneously displace the germane effort involved in retrieving prior knowledge, monitoring strategies, and correcting errors, thereby producing the "learning penalty" observed when students who excel during GPT-supported practice later perform worse on independent exams (Bastani et al., 2025). Teacher and student reports of high error rates and a readymade mentality, in which learners use AI to obtain finished answers without engaging deeply with the underlying concepts, echo concerns about such over-reliance. Correlational evidence suggests that adolescents with greater executive functioning difficulties are especially likely to view GenAI as useful for completing schoolwork; Researchers warn that this may encourage the substitution of AI for their own planning and problem-solving over time (Ergene & Ergene, 2025; Klarin et al., 2024).

These behavioral patterns find a potential neurocognitive parallel in recent work by Kosmyna et al. (2025). They examined essay writing among adults and reported that repeated use of an LLM assistant was associated with reduced functional connectivity in brain networks implicated in higher order cognition, compared with unaided writing. The authors describe this pattern as "cognitive debt," using it as a metaphor to suggest that offloading effort in the moment may reduce opportunities to practice retrieval, synthesis, and organization that support skill consolidation. However, this evidence is limited to a specific adult writing context and does not involve K-12 students. Accordingly, in this review we treat the cognitive debt account as a theoretical lens rather than direct empirical evidence of developmental risk in K-12 learners. Nevertheless, this theoretical mechanism aligns with a recent systematic review of 14 empirical studies, indicating that students increasingly favor fast, optimal AI-generated solutions over slower, practical problem-solving, raising concerns that over-reliance on AI tools may erode critical cognitive skills (Zhai et al., 2024). At most, it offers a plausible mechanism that future studies should test with child and adolescent samples and in authentic classroom tasks, including whether any neurocognitive effects persist after AI support is removed and whether they relate to the learning outcomes reported in the K-12 literature.

At the same time, the available evidence suggests that GenAI's impact on learning is contingent rather than uniformly detrimental. A meta-analysis of AI-chatbot-based interventions reported an overall positive effect on student learning performance, with particularly strong gains when AI is embedded in teacher-supported activities rather than used in isolation (Wu & Yu, 2024). Complementing this, Xia et al. (2023) showed in a university sample that students with stronger prior English knowledge and higher perceived autonomy and competence demonstrate more effective self-regulated learning with GenAI, whereas those with weaker language skills are less able to benefit and more vulnerable to unproductive use patterns. Taken together with the learning penalty observed when GenAI replaces rather than supports core practice (Bastani et al., 2025), these findings suggest that the risk of harmful cognitive offloading is greatest when AI is allowed to substitute for foundational skill acquisition, whereas GenAI is more likely to function as a productive scaffold when it is deliberately positioned to amplify existing competencies within well-designed, instructionally supported tasks.

4.3. Intellectual agency: creativity, authenticity, and identity development

The creativity-related risks identified in this review extend well beyond narrow concerns about plagiarism or task substitution and cut to the heart of how students come to see themselves as authors and creators. In primary school writing classrooms, students' frustration with biased or ill-fitting AI-generated images (e.g., stereotyped depictions of "poor families") shows how GenAI can both constrain children's imaginative space and implicitly communicate whose stories and bodies are treated as normal or worth representing (LaMear & von Gillern, 2025). In secondary literacy settings, students described AI-generated text as technically polished but "soulless" or "not really mine," with some expressing moral unease about submitting such work; in doing so, they reframed originality as a question of voice and ownership rather than mere compliance with plagiarism rules (Higgs & Stornaiuolo, 2024).

Viewed through an Eriksonian lens, these findings are developmentally consequential. For example, adolescence is a period in which they experiment with roles and narratives to construct a coherent sense of self, in part through authentic expression in valued domains such as writing, art, and design (Erikson, 1968). When GenAI becomes the de facto author of academic and creative products, it risks displacing precisely those acts of struggle, revision, and personal risk-taking that undergird the feeling that "this is me." Reports from teachers that heavy GenAI use dampens students' originality and their own creative agency, especially when outputs are copied wholesale or only superficially edited, reinforce this concern (López Santos et al., 2025). At the same time, students' emerging critical awareness of AI bias and their efforts to preserve an authentic voice suggest that, when engaged explicitly and reflexively, GenAI can also serve as a site for developing sociotechnical and ethical literacy (Higgs & Stornaiuolo, 2024; LaMear & von Gillern, 2025). Taken together, this review reframes creativity risks as not only about "cheating," but about how GenAI may either erode or enrich identity-relevant practices depending on how it is framed, constrained, and discussed within learning environments.

4.4. Ecological education system

While this review identifies general risks associated with GenAI, these harms are unlikely to be distributed evenly across the student population. Instead, the findings suggest the potential existence of a deeper *digital divide*, where inequality is defined not by access to technology, but by the quality and purpose of its use. As highlighted by Jang and Choi (2025), high-achieving students tend to utilize GenAI as a heuristic scaffold to deepen inquiry, whereas marginalized or lower-achieving students are more likely to employ it as a substitutive tool for task completion. Consequently, the developmental impact of GenAI is heavily contingent upon the student's broader social and material context (UNESCO, 2025). Without robust adult guidance, teacher capacity, and protective school policies, GenAI risks functioning as a "crutch" that displaces student agency rather than a scaffold that supports it, ultimately amplifying existing inequities and calcifying stratified developmental outcomes.

For instance, the risk of a "learning penalty" from cognitive offloading is likely to be most acute in under-resourced schools. In settings with larger class sizes and over-extended teachers, students may turn to GenAI not for supplemental help, but as a primary substitute for unavailable human instruction. Without a teacher to model critical use, enforce process-oriented tasks, and scaffold interaction, cognitive offloading becomes the path of least resistance, potentially cementing learning gaps between students who have access to such mediation and those who do not. The risks to psychological wellbeing may be similarly unequal. Students in schools lacking adequate counseling and mental health services may be more likely to turn to AI chatbots as a first resort, elevating the risk of receiving generic or inappropriate advice and delaying access to professional care, which is already scarce.

The risks to creativity and authenticity are also intertwined with identity and social position. As the work of LaMear and von Gillern (2025) suggests, AI models trained on narrow, majority-culture datasets can produce biased outputs that stereotype or erase students from marginalized backgrounds. This is not merely a technical flaw; it is a developmental one. When a tool consistently fails to represent a child's reality or renders their community in stereotypical ways, it can stifle creative expression and reinforce feelings of marginalization, implicitly teaching them whose stories are considered "normal" or valuable. The pressure to adopt a generic, AI-polished "voice" may be felt more acutely by students whose home languages and cultural communication styles already diverge from dominant academic norms, creating another layer of inauthenticity in their identity work.

4.5. Implications for pedagogy, design, and governance

Our findings suggest that reported risks associated with GenAI are contingent on how tools are used and governed, and therefore may be reduced through instructional, design, and policy choices. The following points synthesize implications that are consistent with the patterns and recommendations reported in the reviewed evidence, while recognizing that the current literature remains early and heterogeneous.

At the classroom level, the evidence points towards three interlocking pedagogical priorities. First, GenAI should be treated explicitly as a scaffolded tool rather than an answer engine: teachers can encourage students to use AI for brainstorming, receiving hints, and critiquing model answers, while requiring them to document reasoning, compare AI suggestions with their own ideas, and reflect on discrepancies. Such developmentally attuned approaches align with emerging frameworks like Tao (2025), which proposes tiered restrictions and objectives based on age-specific cognitive readiness to preserve student agency. Second, assessment practices may need to shift towards process-oriented tasks, such as oral explanations, in-class problem-solving, and portfolios showing drafts and revisions, that are less susceptible to full outsourcing and foreground students' thinking and creative decisions (Bastani et al., 2025; Jang & Choi, 2025). Third, GenAI related activities can be used to foster AI literacy and critical digital citizenship, including discussion of bias, fairness, privacy, and the emotional and identity implications of co-creating with machines (Chan, 2025; Higgs & Stornaiuolo, 2024; LaMear & von Gillern, 2025).

At the tool design level, the contrast between direct answer support and scaffolded hint-based support suggests that interface features may function as practical levers that shape learning behavior (Bastani et al., 2025). Since LLM outputs are generated through probabilistic prediction, systems can also produce hallucinated content and reproduce biases, especially when prompts are underspecified or when uncertainty is hidden from users. This implies that design level mitigation should not rely only on user caution. Instead, designers and procuring institutions can consider features that prompt metacognitive monitoring, surface uncertainty and limitations, and constrain responses in age and subject appropriate ways. Where feasible, developers can also explore training and tuning approaches that incorporate pedagogically grounded interaction patterns, for example datasets that model stepwise hints, ask students to explain reasoning, invite verification, and encourage productive struggle rather than rapid answer delivery. These design implications are forward looking, because most studies in the current K-12 evidence base evaluate tool use rather than model training strategies, but they clarify a pathway for reducing hallucination and bias risks while supporting learning processes.

At the policy and capacity building level, the reviewed literature highlights needs for clear and context sensitive guidance on academic integrity, data privacy, and equity goals in GenAI use (Mustafa et al., 2024). In line with these reports, professional development can be framed as a capacity building condition that enables educators to interpret evolving guidance and implement GenAI in developmentally appropriate ways, positioning teachers as instructional facilitators who structure student engagement with AI rather than as gatekeepers or passive adopters (Giannakos et al., 2025; Lee & Kwon, 2024).

4.6. Limitations and future directions

This scoping review is subject to several limitations. First, the current evidence base represents only the nascent first wave of post-GPT research in K-12 settings. Our search window (2022–2025) yielded 22 short-term studies, primarily concentrated in mathematics, programming, and writing. Consistent with the aim of a scoping review to map the breadth of available literature rather than its quality (Khalil et al., 2021), no formal appraisal of the included studies was conducted. The synthesis, therefore, draws on evidence of varying methodological rigor, and the strength of our conclusions is limited by this inherent heterogeneity. Second, by intentionally focusing on risks to answer our research questions, the resulting map is necessarily risk-oriented and should not be interpreted as a balanced estimate of GenAI's overall impact. This limitation is likely compounded by publication bias, where significant negative effects may be over-represented compared to null or ambiguous findings.

These limitations highlight urgent priorities for future investigation. Specifically, most studies focus on single age bands or fail to disaggregate results by grade level. Future research can consider rigorously investigating how risks differ between primary learners and secondary learners. Additionally, longitudinal and experimental designs are essential to trace how GenAI use shapes trajectories of executive functioning, self-regulation, mental health, and creative identity over time.

Finally, future scholarship should address equity and design. Studies should examine moderators such as baseline cognitive skills, socioeconomic status, and access to guidance to determine if differential prompting skills or infrastructure widen existing inequalities. These contrasts motivate a testable hypothesis for K-12 education: that constrained, domain-specific AI offering metacognitive prompts may be more developmentally appropriate than highly capable, general-purpose LLMs.

5. Conclusion

This scoping review synthesizes empirical evidence regarding the potential risks of integrating GenAI into K–12 education. Analysis of 22 studies reveals that risks primarily cluster around psychological well-being, intellectual agency and the educational ecosystem. In the cognitive domain, cognitive offloading and over-reliance on algorithmic outputs can degrade independent problem-solving skills and distort metacognitive self-assessment. Furthermore, the integration of these tools may erode creative authorship, as students struggle to maintain authentic voice and ownership over their work in the face of algorithmic perfectionism.

Beyond individual impacts, this review identifies systemic risks relating to equity and institutional preparedness. A deeper digital divide may emerge, in which the risks of AI dependency are distributed unevenly, potentially widening the achievement gap between students who use GenAI to facilitate deeper learning and those who use it as a substitute for effort. These challenges are exacerbated by a lack of clear governance and privacy concerns, as well as the potential for AI to replace essential human interactions that are fundamental to psychosocial development during childhood and adolescence.

Ultimately, safely integrating GenAI requires shifting the educational focus from rapid technical adoption to ecological protection. The evidence indicates that without intentional, developmentally responsive safeguards, GenAI could function as a crutch rather than a tool for empowerment. Future educational practice should therefore prioritize pedagogical strategies that value process over product and enforce human-in-the-loop learning. By establishing rigorous governance and fostering critical AI literacy, the K–12 sector can ensure that these powerful technologies support, rather than displace, the productive struggle and human agency that remain central to the learning process.

CRediT authorship contribution statement

Sisi Tao: Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Min Lan:** Writing – review & editing. **Minjuan Wang:** Writing – review & editing. **Hui Li:** Writing – review & editing.

Ethics statement

The study does not involve human participants.

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Declaration of competing interest

None.

Appendix A. Supplementary data

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Data availability

This study does not contain any empirical data.

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